

Deep Learning-Based Detection of Abnormal Signals in Spectrogram Data Using ResNet18

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Abstract—Wireless communication environments are increasingly vulnerable to abnormal and potentially malicious signal activities, including interference, unauthorized transmissions, and spectrum misuse. Detecting such anomalous behaviors is an important problem in wireless security and spectrum monitoring. In this project, a deep learning-based approach is proposed for spectrum anomaly detection using spectrogram representations of wireless signals. The experiments are conducted on the Wireless Anomaly Signal Dataset, which combines measured LTE/5G signals with synthetic anomaly overlays including tone, chirp, and pulse patterns. A pretrained ResNet18 model [1] is fine-tuned to classify spectrogram samples into four categories: tone, chirp, pulse, and normal. Experimental results show that the proposed model achieves a validation accuracy of 96.48% and maintains strong robustness across multiple signal-to-noise ratio (SNR) conditions, reaching up to 98.01% test accuracy at higher SNR values. These results demonstrate that transfer learning with convolutional neural networks is an effective strategy for spectrum-level anomaly detection in wireless security applications.

Index Terms—anomaly detection, spectrogram classification, ResNet18, deep learning, wireless signal security, transfer learning

I. INTRODUCTION

Wireless communication systems are becoming increasingly complex due to the growing use of LTE, 5G, and shared spectrum technologies. As spectrum environments become more crowded, they also become more vulnerable to abnormal or suspicious signal activity such as interference, unauthorized emissions, spectrum misuse, and unexpected waveform behavior. Detecting such anomalies is therefore an important problem in wireless communication security and spectrum monitoring.

Traditional signal anomaly detection methods often rely on handcrafted features, threshold-based rules, or signal-processing techniques designed for specific environments. Although these methods can perform well in controlled settings, they may struggle when signal abnormalities vary in shape, duration, or noise conditions. Recent work has shown that deep learning methods can offer more flexible and robust alternatives for spectrum analysis and abnormal signal identification [2], [3].

A particularly effective strategy is to represent wireless signals in the time–frequency domain using spectrograms,

allowing the anomaly detection problem to be treated as an image classification task. This makes it possible to leverage pretrained convolutional neural networks (CNNs) originally developed for computer vision. In this project, a pretrained ResNet18 model [1] is fine-tuned to detect and classify abnormal wireless signal patterns from spectrogram data.

The experiments are conducted using the Wireless Anomaly Signal Dataset, which was specifically designed for spectrum-level anomaly detection and combines measured LTE/5G signals with synthetic anomaly overlays such as tone, chirp, and pulse patterns [4]. From a security perspective, these anomalies can be interpreted as suspicious or non-standard spectrum behaviors that may indicate interference or unauthorized activity.

The main contributions of this work are as follows:

- Development of a deep learning pipeline for spectrogram-based wireless anomaly detection.
- Fine-tuning of a pretrained ResNet18 model for multiclass classification of normal and anomalous signal patterns.
- Evaluation of model performance under varying signal-to-noise ratio (SNR) conditions.
- Analysis of the applicability of deep learning to spectrum monitoring in a wireless security context.

II. RELATED WORK

Spectrum monitoring and anomaly detection have attracted increasing attention in recent years due to the rapid growth of wireless communication systems and the need for secure and reliable spectrum usage. Traditional approaches often rely on signal energy detection, cyclostationary feature analysis, or handcrafted statistical features. While such methods can be effective in constrained scenarios, they often require substantial domain knowledge and may generalize poorly to complex or noisy environments.

Recent advances in deep learning have enabled new approaches for wireless signal analysis by treating spectrum data as image-like or structured representations. Bhatti *et al.* [2] demonstrated that deep learning can be effectively applied to shared spectrum monitoring, showing the potential of CNN-based approaches for automated wireless signal characterization. Their work supports the use of learned visual features rather than manually engineered signal descriptors.

More recently, Kim *et al.* [3] investigated spectrum anomaly detection using deep neural networks from a wireless signal

perspective. Their work highlights the effectiveness of deep models for distinguishing anomalous spectrum activity under realistic signal conditions. In a related study, Kim *et al.* [5] explored anomaly detection for wireless cellular communication using synthetic anomaly injection, demonstrating that controlled anomaly generation can be used to construct useful benchmark scenarios for supervised learning.

This project is also closely connected to the Wireless Anomaly Signal Dataset, introduced in [4]. The dataset provides a realistic benchmark for spectrum-level anomaly detection by combining measured LTE/5G signals with synthetic anomaly overlays. Compared with earlier datasets that are either overly simplified or limited in anomaly diversity, WASD provides a practical foundation for evaluating deep learning-based detection models.

In contrast to prior work that may focus on detection alone or use custom architectures, this project investigates the use of a pretrained ResNet18 model as a transfer learning solution for anomaly classification in spectrogram data. This choice is motivated by the strong feature extraction capability of residual networks and their efficiency for medium-scale image-based classification tasks.

III. DATASET AND PROBLEM FORMULATION

A. Dataset Overview

The experiments in this project are based on the Wireless Anomaly Signal Dataset, a dataset designed for spectrum-level anomaly detection in fixed urban wireless environments [4]. The dataset combines measured LTE/5G signals with synthetic anomaly overlays to support detection and localization tasks.

The anomaly types considered in this project are: Tone, Chirp, Pulse, or Normal otherwise.

These classes were used to define a four-class classification problem, where the objective is to identify whether a given spectrogram corresponds to normal wireless activity or to one of several abnormal signal patterns.

B. Training and Validation Data

The main dataset used for training and validation contains 4500 base samples across 19 frequency bands, resulting in a total of 85,500 spectrogram samples. The signal data are stored in `Abnormal_spectrogram.npy`, while the corresponding labels are provided in `Spectrum_label.csv`.

For model training, the labels were mapped as follows:

- tone $\rightarrow$ class 0
- chirp $\rightarrow$ class 1
- pulse $\rightarrow$ class 2
- normal $\rightarrow$ class 3

C. Test Data and SNR Conditions

To evaluate robustness under noisy conditions, the model was tested on multiple spectrogram files corresponding to different signal-to-noise ratio (SNR) values. Each test file contains 500 base samples across 19 bands, yielding 9,500 spectrogram images per file. The evaluated SNR values range from -10 dB to 10 dB.

D. Security Interpretation

Although the anomaly classes in dataset are synthetically overlaid signal patterns, they are meaningful from a wireless security perspective. In practice, such anomalies may represent suspicious transmissions, unintended interference, non-standard emissions, or spectrum misuse. Therefore, the classification task can be interpreted as a simplified anomaly detection mechanism for identifying potentially problematic wireless activity.

IV. METHODOLOGY

A. Spectrogram-Based Representation

The anomaly detection task was formulated as an image classification problem by representing wireless signal activity in spectrogram form. Spectrograms capture the time-frequency characteristics of a signal and provide structured visual patterns that are well suited for convolutional neural networks.

This representation is particularly useful because anomalies such as tones, chirps, and pulses often exhibit distinct visual signatures in the spectrogram domain.

B. Model Architecture

This project uses ResNet18 as the classification model. ResNet18 is a residual convolutional neural network that introduces skip connections to improve optimization and enable efficient feature learning in deep architectures. It was selected for this work because it provides a good balance between classification performance and computational efficiency.

A pretrained ResNet18 model was used as the backbone network, and its final fully connected layer was replaced to output four classes corresponding to the target labels: tone, chirp, pulse, and normal.

C. Data Preparation

The spectrogram data were converted into image-like tensors suitable for input into the ResNet18 architecture. Since the pretrained model expects three-channel inputs normalized similarly to ImageNet data, preprocessing was applied to adapt the spectrogram representations to the expected format.

The preprocessing pipeline included:

- loading the spectrogram arrays and labels,
- converting spectrogram samples into image-compatible tensors,
- applying normalization for transfer learning,
- splitting the dataset into training and validation sets.

D. Training Setup

The model was fine-tuned using supervised learning on the WASD dataset. The classification objective was optimized using cross-entropy loss. Due to the relatively large size of the dataset and the associated computational cost, the training process in this project was limited to two epochs. Despite this constraint, the model achieved strong performance, indicating that the pretrained ResNet18 backbone was able to adapt

effectively to the spectrogram classification task. Performance was evaluated using standard classification metrics.

The evaluation metrics used in this project include:

- accuracy,
- precision,
- recall,
- F1-score,
- confusion matrix.

V. EXPERIMENTAL RESULTS

A. Training and Validation Performance

The model converged rapidly during training and achieved strong validation performance after only two epochs. This is particularly noteworthy given that the training duration was intentionally limited due to the relatively large dataset size and computational cost.

Table I summarizes the training and validation metrics.

TABLE I
TRAINING AND VALIDATION PERFORMANCE

Epoch	Train Loss	Train Acc.	Val Loss	Val Acc.
1	0.1875	0.9290	0.1227	0.9510
2	0.1083	0.9600	0.0923	0.9648

The best validation accuracy obtained was 96.48%, indicating that the pretrained ResNet18 model successfully adapted to the spectrogram anomaly classification task.

B. Validation Classification Report

Table II presents the class-level validation performance.

TABLE II
VALIDATION CLASSIFICATION REPORT

Class	Precision	Recall	F1-score
Tone	0.9802	0.9915	0.9858
Chirp	0.9726	0.9609	0.9667
Pulse	0.9565	0.9547	0.9556
Normal	0.9570	0.9584	0.9577

The model achieved balanced performance across all classes, with the highest performance observed for the tone class. This suggests that tone anomalies exhibit particularly distinctive time–frequency characteristics in spectrogram form.

C. Confusion Analysis

The confusion matrix indicates that most samples were correctly classified. The most noticeable classification ambiguity occurred between pulse anomalies and normal samples, which is reasonable because some pulse-like distortions may overlap visually with standard signal structures in the spectrogram domain.

Figure 1 provides a visual summary of class-level prediction behavior on the validation set.

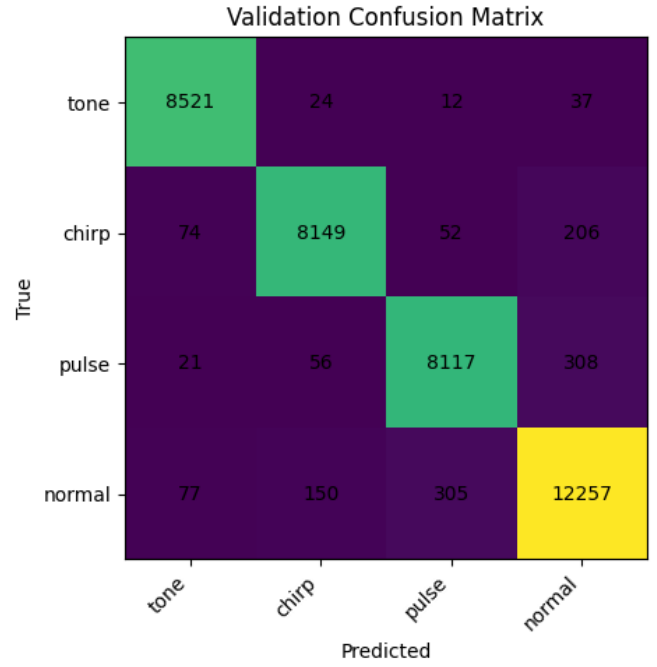


Fig. 1. Confusion matrix on the validation set. Most samples are correctly classified, with minor confusion observed mainly between pulse anomalies and normal signals.

TABLE III
TEST PERFORMANCE ACROSS SNR LEVELS

SNR (dB)	Accuracy	Macro F1
-10	0.8760	0.8598
-6	0.9507	0.9464
-4	0.9635	0.9601
-2	0.9691	0.9663
0	0.9707	0.9686
2	0.9745	0.9731
4	0.9776	0.9763
6	0.9753	0.9739
8	0.9780	0.9766
10	0.9801	0.9789

VI. ROBUSTNESS UNDER DIFFERENT SNR CONDITIONS

A key objective of this project was to evaluate whether the model remains effective under noisy spectrum conditions. To this end, the trained classifier was tested across multiple signal-to-noise ratio (SNR) levels.

The results show a clear relationship between signal quality and classification performance. The model performs worst at -10 dB, where noise significantly obscures the spectrogram structure, but improves steadily as the SNR increases. At moderate and high SNR levels, the classifier remains highly stable and consistently achieves strong performance.

From a security perspective, these results are important because they indicate that the proposed detection approach remains useful even under degraded signal conditions, although very high noise levels can reduce anomaly distinguishability.

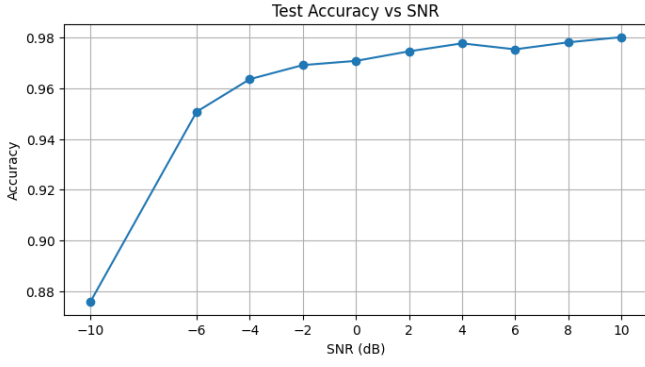


Fig. 2. Test accuracy across different signal-to-noise ratio (SNR) levels. Model performance improves as signal quality increases and remains robust across most tested conditions.

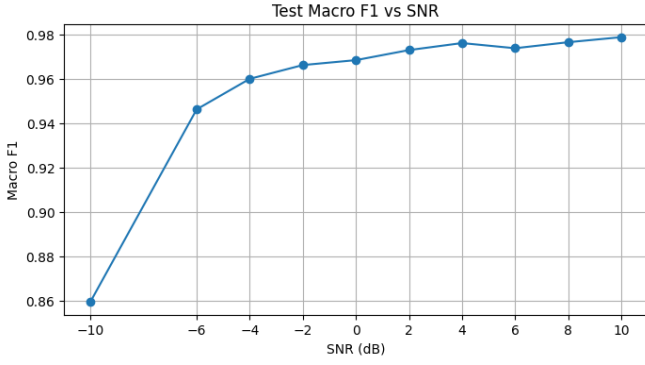


Fig. 3. Macro F1-score across different SNR levels. The results show stable class-balanced performance except under the lowest noise conditions.

VII. DISCUSSION

The results of this project demonstrate that transfer learning with ResNet18 is an effective approach for spectrum anomaly detection in wireless environments. Several important observations can be drawn from the experiments.

First, the model achieved strong performance after only a small number of training epochs, suggesting that pretrained visual features transfer well to spectrogram-based signal analysis. This supports prior work showing the usefulness of deep learning in spectrum monitoring and anomaly detection [2], [3].

Second, the class-level results indicate that some anomaly types are easier to distinguish than others. Tone and chirp anomalies achieved particularly strong classification performance, likely because they produce more distinctive and stable visual structures in the spectrogram domain. In contrast, pulse anomalies showed slightly greater confusion with the normal class, which may be due to partial overlap in spectral appearance.

Third, the SNR experiments confirm that model performance degrades under severe noise conditions but remains robust across most of the tested range. This is particularly relevant for wireless security applications, where real-world

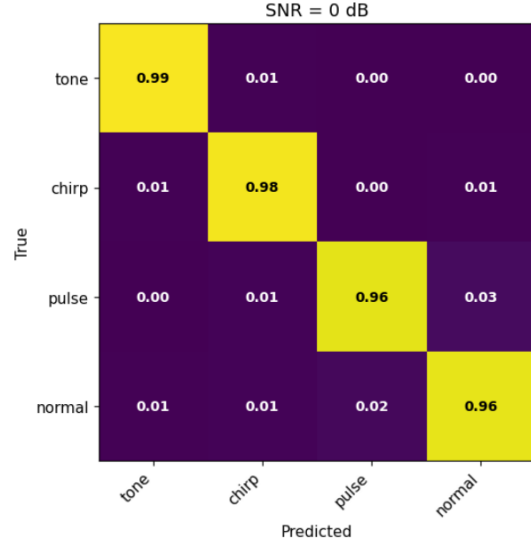


Fig. 4. Normalized confusion matrix for the test set at 0 dB SNR. The classifier maintains strong performance across all classes, with only minor confusion observed mainly between pulse and normal samples.

spectrum monitoring systems must operate in noisy and unpredictable environments.

Despite these promising results, this work has several limitations. Only one model architecture was evaluated, and the training duration was limited to two epochs due to the relatively large dataset size and computational constraints of the project environment. In addition, the experiments were conducted on a controlled benchmark dataset rather than a fully operational real-world deployment scenario.

VIII. FUTURE WORK

Future work may include extending the training process beyond two epochs to better explore the full learning capacity of the model. Due to the relatively large dataset size and computational cost, the current training setup was intentionally limited, and a longer training schedule may further improve model generalization. It would also be valuable to evaluate alternative deep learning architectures such as EfficientNet, MobileNet, and object detection models such as YOLO. In particular, YOLO could be explored to detect and localize anomalous signal patterns directly within spectrogram images, rather than only performing whole-image classification. This could be especially useful in cases where the anomaly occupies only a limited region of the time–frequency representation. Additional future directions include applying data augmentation, investigating adversarial robustness, and extending the model toward real-time anomaly detection in live wireless monitoring systems.

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